Department of Computer Science and Engineering

Program: BE

Date	29 April 2025	Maximum Marks	50
Course Code	**CS362AI**	Duration	90 min
6th Sem	VI Semester	CIE-I	
Network Programming and Security			

	QUIZ	M	L	CO
1	Referring the below transition state diagram, answer 1 and 2 Client Server | -------- FIN ----------> | | | | <-------- ACK ---------- | | | | (server app still running) | | | <-------- FIN ---------- | | -------- ACK ----------> | What TCP state is the client in after its FIN is acknowledged but before receiving server's FIN?	1	L3	CO1
2	What state does the client enter after sending the final ACK?	1	L3	CO1
3	Referring the below transition state diagram, answer 3 and 4 Client Server | -------- SYN ----------> | | <-------- SYN-ACK -------- | | -------- ACK ----------> | | (Client app crashes) | What state is the server in after the client's crash?	1	L3	CO1
4	Will the server socket close automatically if no FIN is received?	1	L3	CO3
5	Referring to the following output of the netstat command, answer 5 and 6 **netstat output on server:** Proto Recv-Q Send-Q Local Address Foreign Address State tcp 0 0 192.168.1.100:8080 192.168.1.10:52345 TIME_WAIT Why is this socket in TIME_WAIT state?	1	L3	CO2
6	Can the server rebind to port 8080 immediately? What's the condition?	1	L3	CO3
7	A concurrent TCP server uses fork() to handle clients. After the server accepts a connection and forks, which socket(s) must be closed in the child and parent processes?	2	L3	CO3
8	After a client successfully connects to a server using connect(), what is the role of the getpeername() function?	2	L3	CO2

Sl. No.	Test Questions	M	L	CO
1a	With a neat diagram explain the TCP connection establishment and termination.	05	L1	CO1
1b	The **sequence number field** in the TCP header is 32 bits long, allowing it to represent over **4 billion bytes** of data. While this may seem sufficient, is it really enough when transferring large amounts of data over a long-lived or high-speed connection? Additionally, since the **initial sequence number (ISN)** is chosen randomly, it may start close to the maximum value (2^{32} - 1). In such cases, wouldn't the sequence number **wrap around** to 0 much sooner? How does TCP handle situations where more than 4 GB of data is transmitted, or when sequence numbers wrap around quickly due to a high ISN?	05	L3	CO1
2a	Why is the TIME_WAIT in TCP typically set to 2 RTT (Round-Trip Time)? What would happen if the timeout was set to less than 2 RTT or more than 2 RTT?	05	L4	CO2
2b	When TCP sends a **SYN** or **FIN** with **Sequence Number = x**, the corresponding **ACK** is **Acknowledgment Number = x + 1**, meaning that SYN and FIN each consume one unit in the sequence number space. Is this necessary? If so, give an example of the ambiguity that would arise if the acknowledgment were **x** instead of **x + 1**. If not, explain why.	05	L4	CO2
3a	Explain the various system calls used by a server to establish and manage a socket connection in a TCP/IP network communication.	05	L1	CO3
3b	Demonstrate the working of the exec system call by creating a child process that executes a different program, passing arguments to the *main() function* of the new program.	05	L4	CO2
4a	In a TCP client-server application, both getsockname() and getpeername() functions are called on a connected socket (connectfd) in the child process. Explain what each function returns, and under what circumstances the two might return the same IP address and port number. Justify your answer with a brief explanation.	05	L3	CO2
4b	In a concurrent server application, when a child process is created, both the listening socket (listenfd) and the connected client socket (connectfd) are duplicated in the child process. Typically, the parent process closes connectfd, while the child process closes listenfd. According to the TCP state transition diagram, closing a socket should trigger a FIN signal and initiate connection termination. Given this behavior, does closing a socket in a concurrent server application immediately terminate the TCP connection? Provide a proper justification for your answer.	05	L3	CO1
5b	Write a program for TCP Echo server and TCP Echo client	10	L1	CO3